# CrateTag

Find the release year and genre for your MP3s and write them safely. Version 0.12.0 for macOS.

CrateTag changes only the artist, title, year and genre tags, and only when you ask it to. Serato's cue points, loops, colors, beatgrids and waveform data stored inside each MP3 are left byte-for-byte intact, and the audio is never re-encoded.

> **Yellow boxes** are the commands to type in Terminal the first time you set CrateTag up. Open Terminal from Applications > Utilities, paste a command, and press Return. Gray boxes are optional or for later.

## 1. Install CrateTag

The installer checks for everything CrateTag needs and installs whatever is missing, then builds CrateTag and puts it in your Applications folder.

1. Download **CrateTag.zip** and double-click it to unzip it in your Downloads folder.
2. Open **Terminal** (Applications > Utilities > Terminal), paste this command, and press Return:

**RUN IN TERMINAL (FIRST-TIME SETUP)**

```
bash ~/Downloads/CrateTag/"Install CrateTag.command"
```

What happens next:

- It checks for **Python 3.10 or newer**. If it's already installed, nothing else is installed.
- If Python is missing, it installs **Homebrew** (if you don't have it) and then **Python**. macOS asks for your Mac password; nothing appears as you type, which is normal. A window may also offer to install Apple's **Command Line Tools**: click Install. This part can take 5 to 15 minutes.
- It then builds CrateTag (a minute or two), installs it in **Applications**, and opens it. The Terminal window closes by itself when it's done; if anything goes wrong it stays open with the reason.

If you unzipped CrateTag somewhere other than Downloads, change the path in the command to match. The command is run with bash, so macOS's "downloaded from the internet" block doesn't stop it.

### Installing the pieces yourself (optional)

Prefer to do it by hand? These are the same steps the installer takes.

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"
brew install python
cd ~/Downloads/CrateTag
./build_mac.sh --test --install
```

Skip the first three lines if Homebrew is already installed (check with brew --version). On older Intel Macs the two eval lines aren't needed. To run CrateTag without building the app, use ./run_from_source.sh instead of the last line.

## 2. Connect data sources (Settings)

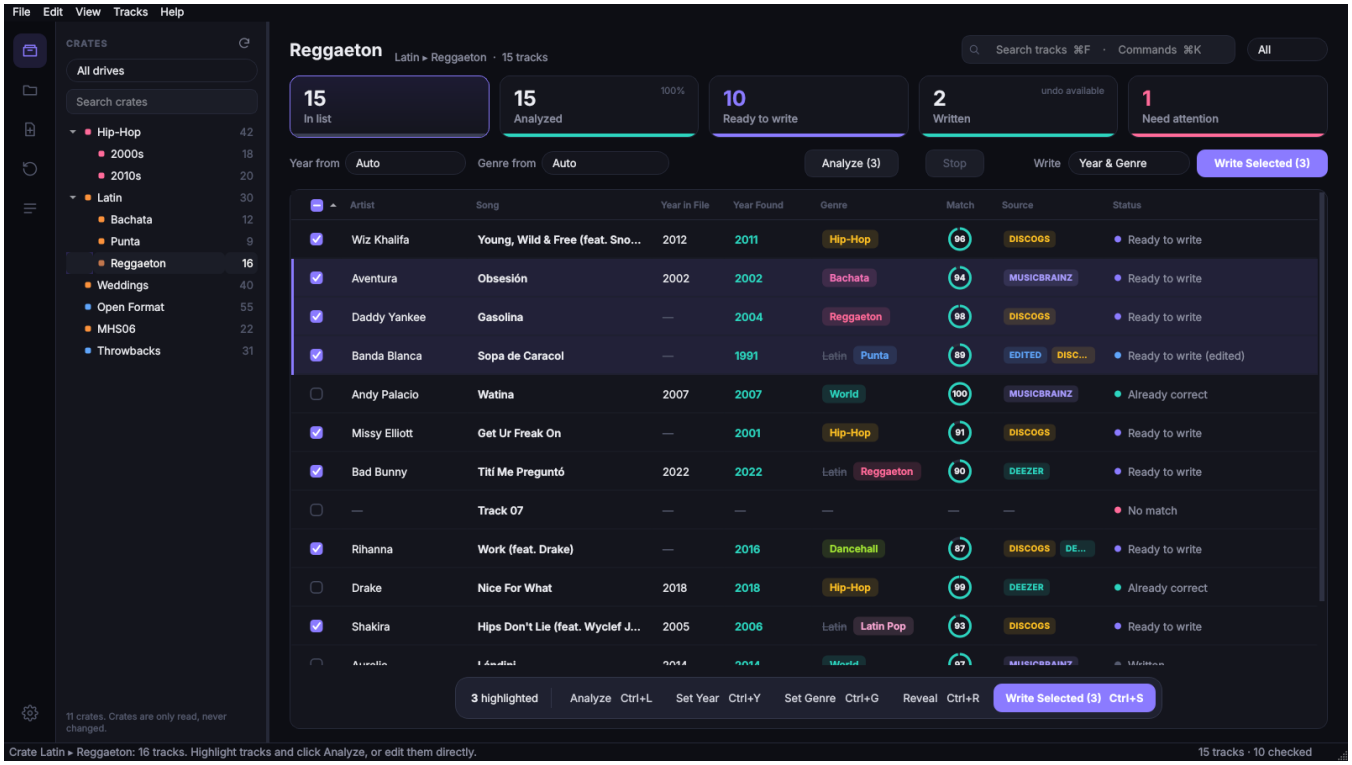
- **MusicBrainz** is on by default. Free, no account. Good for release years.
- **Deezer** is on by default. Free, no account or key. Strong on release years for recent hits and new releases; its genres are broad (Rock, Dance, Hip Hop, Latin) and are used when the others have none.
- **Apple iTunes** is on by default as a backup. Free, no account or key. Apple allows only about 20 searches a minute, so CrateTag asks it only when the other sources leave a track's year or genre empty (or when you pick Apple iTunes under Year from or Genre from). Its genres are broad, like Hip-Hop/Rap, Latin or Pop, and compilation dates are ignored. Turn it off in Settings if you like.
- **Discogs** (strongly recommended) has the best genres and specific styles. Create a free account at discogs.com, open Settings > Developers, click **Generate new token**, and paste it into CrateTag's Settings.
- **AcoustID** (optional) identifies songs by sound, for files named things like Track 07.mp3. It needs a free API key from acoustid.org pasted into Settings, plus a fingerprint tool. The installer adds the tool for you with this command:

```
cd ~/Downloads/CrateTag
./install_deps.sh --acoustid
```

## 3. Using CrateTag

1. **Pick a Serato crate** in the list on the left (see Working with Serato crates), or click **Add Folder**, or drag a folder onto the window. Tags are read; nothing is changed.
2. **Highlight the tracks** you want (click, Shift-click a range, Command-click to add), or use the check-all box at the top of the checkbox column. Click **Analyze**. The button shows how many tracks it will analyze.
3. **Review.** Year Found shows the year analysis found, in green (it stays empty until a track is analyzed); compare it with Year in File. Genre shows the file's genre crossed out next to the new one. Confident matches are checked automatically.
4. Choose what to write with **Write** above the track list: **Year & Genre**, **Year only** or **Genre only**. The value that won't be written is grayed out.
5. **Write**, then confirm. With tracks highlighted the button reads **Write Selected** and writes just those, no checkboxes needed. With nothing highlighted, or if the highlighted tracks have nothing to write, it reads **Write Checked** and writes the checked tracks. Click the empty space below the list to clear the highlight.
6. **In Serato**, select the updated tracks and use **Rescan ID3 Tags** so the library shows the new values.

## The CrateTag window




CrateTag 0.10 with a Serato crate loaded and analyzed.

- **Left strip:** Crates, Folders, Add Files, Undo a Previous Write, Activity Log, and Settings at the bottom. Click Crates or Folders again to hide that column and give the track list more room.
- **Crates:** your Serato crates with their track counts. Each top-level crate gets its own color, shared by its sub-crates; the colors are only to help you find them. The drive picker at the top chooses which drive's crates to show.
- **Folders:** browse your Music folder, home folder and drives. Double-click a folder (or click **Load This Folder**) to load its MP3s, including subfolders. Right-click for Add Folder to List.
- **The five counts across the top** are filters: **In list**, **Analyzed**, **Ready to write**, **Written** and **Need attention** (no match, errors, weak matches, or a missing year or genre). Click one to show only those tracks, click it again (or In list) to show everything. The check-all box, Analyze and Write then work on what's showing.
- **Track list:** Artist, Song, Year in File, Year Found (only what analysis found, in green; blue for a year you typed) and Genre (old crossed out, new as a colored tag), Match (a ring with the match %), Source (which service found it, or EDITED for your own changes) and Status.
- **Buttons above the list:** Year from and Genre from on the left, Analyze and Stop under the Ready to write and Written counts, and on the right what to write and the Write button. Clicking another crate replaces the list (CrateTag asks first if you have changes that aren't written yet); to empty the list yourself, use File > Clear.
- **Check-all box:** the box at the top of the checkbox column checks every track showing (the whole crate, or just your search or filter results). Click it again to uncheck them all. A dash means some are checked. Clicking elsewhere in that column's header still sorts by checked tracks. Clicking a track's own checkbox only checks or unchecks it; it doesn't highlight the row.

- **Floating bar:** highlight tracks and a bar appears at the bottom with Analyze, Set Year, Set Genre, Reveal and Write (the list makes room for it, so no track is hidden underneath), with their shortcuts.
- **Command palette:** press Command-K, type part of any command (for example "reveal" or "set year") and press Return.
- **Prefer the old look?** Choose View > Classic Look, then reopen CrateTag. Choose it again to come back.

### Searching tracks

- Type in the **search box above the track list** (or press Command-F) to show only matching tracks. The menu next to it searches **All** (artist, title and file name), **Artist** only, or **Title** only.
- Case and accents don't matter ("obsesion" finds "Obsesión"), and several words can be in any order ("small things" finds "All The Small Things").
- While searching, the check-all box and **Analyze** cover only the tracks that are showing. The count at the bottom right shows how many tracks are showing out of the total. Clear the box to see everything again.
- The search stays in place when you switch crates, so you can look for an artist across crates one at a time.

### All Songs: your whole Serato library

- **All Songs** at the top of the crates list shows every MP3 in your Serato library, like Serato's **All**. With All drives chosen it combines your Mac and your DJ drive; pick one drive to see just its songs.
- It loads in seconds even for 80,000+ tracks, because it reads Serato's own library file instead of opening every MP3. Videos, WAV and AIFF files aren't listed, since CrateTag edits MP3s only.
- The Year and Genre shown come from Serato's copy of each song's tags. When you analyze a track, CrateTag first reads the real file, so the comparison is always against what's actually in the MP3.
- Use the **track search** to narrow it down (searching all 80,000 tracks takes a fraction of a second), then highlight and Analyze in batches. Analyzing more than 500 tracks at once asks first and shows how long it will take, since MusicBrainz allows about one request per second.
- Highlighting the entire library at once takes a few seconds; highlighting search results is instant.

### Working with Serato crates

The **Serato Crates** list on the left shows your crates the way Serato does: one combined, nested list from your Mac and every connected drive that has a _Serato_ folder, in your Serato crate order.

- **Choose a drive** with the menu above the list: **All drives** (combined, like Serato), your external drive, or this Mac. Your choice is remembered.
- **Search crates** with the box above the list (or press Command-Option-F). Matching crates stay visible with their parent crates opened; clear the box to see everything again.
- Crates shown in grey have no songs of their own on the chosen drive; they just hold sub-crates. Clicking one loads the songs from its sub-crates. Hover over any crate to see which drive it's stored on.
- **Click a crate** to load its songs into the table, in crate order. Then analyze, edit and write as usual.
- Right-click a crate for **Load Crate with Sub-crates**, **Add Crate to List** (adds to what's already loaded), **Reveal Crate File in Finder**, and **Expand All** / **Collapse All**. The number next to each crate is how many songs it holds.

- **External drive:** plug it in and the list updates by itself (or click **Refresh**). The first time, macOS asks to let CrateTag access files on a removable volume. Click **Allow**.
- Songs that can't be found (usually because their drive isn't connected) show **File not found**. Songs that aren't MP3s (M4A, WAV, FLAC) are listed but skipped, since CrateTag edits MP3s only.
- **Smart Crates** appear in their usual place, greyed out and marked (Smart Crate). Serato builds them from rules, so they don't store a song list CrateTag can read. Load the regular crate the songs are in instead.
- If you have unsaved edits and click another crate, CrateTag asks before dropping them.
- **Crates are only read, never changed.** Your crates, cue points and Serato library stay exactly as they are. After writing, select the crate in Serato and use **Rescan ID3 Tags**.
- The **Crates** button in the left strip hides or shows the list. Drag the divider to resize it.

### Choosing the source

If a year or genre looks wrong, change Year from or Genre from above the track list. The table updates immediately from the matches already found, without analyzing again:

- **Year from:** Auto (earliest year among the best matches from any source), MusicBrainz, Discogs, Deezer, or Apple iTunes.
- **Genre from:** Auto (Discogs style, then Discogs genre, then MusicBrainz, then Deezer, then Apple iTunes), Discogs style (specific, such as Trap or Punta), Discogs genre (broad, such as Hip-Hop or Latin), MusicBrainz, Deezer, or Apple iTunes.
- **Auto** fills in from whichever source has an answer. Picking a **specific source** uses only that source, so a blank means that source found nothing. Switch back to Auto, or use Edit Search on that track.
- Each analysis collects matches from every source that's turned on, so switching sources is instant.
- The **Source** column shows where each value came from, for example "Year: MusicBrainz · Genre: Discogs".
- For one track, double-click its row to see every match found and pick the right one, or type a value directly into New Year or New Genre. Values you choose by hand are kept when you change sources.

### Editing artist, title, year and genre by hand

You can change these four fields yourself, with or without analyzing, for one track or many at once:

- **One track:** double-click its Artist, Song, Year Found or Genre cell, type, and press Return. If you hide the Artist column, the artist shows in smaller text under each song; double-click that line to change it.
- **Many tracks, like Serato:** highlight them, click the cell you want to change on one of the highlighted rows, type the value and press Return. CrateTag asks whether to apply it to all highlighted tracks.
- **From the menu:** right-click highlighted tracks and choose **Set Artist…**, **Set Year…** or **Set Genre…** (also in the Tracks menu; Command-Y and Command-G for year and genre).
- Unsaved edits show in **bold blue**; hover over one to see what's in the file now. Edited tracks are checked and marked **Ready to write (edited)**. Click **Write** to save. The undo log covers these edits too.
- Fixed an artist or title? Click Analyze and the search uses your corrected spelling.
- If Write is set to Year only and you change a genre (or the reverse), it switches to Year & Genre so your edit is saved.

### Showing, hiding and arranging columns

- Right-click any column header for a checklist of columns. Uncheck one to hide it, check it to bring it back.
- Drag a column header left or right to move it. Drag the divider between two headers to resize a column, or double-click the divider to fit the column to its contents.
- Your layout is remembered the next time you open CrateTag. **Reset Column Layout** in the same menu puts everything back.
- Year Found and Genre show the value found next to what's in the file, so keep those visible when reviewing. Shown by default: Artist, Song, Year in File, Year Found, Genre, Match, Source and Status. Hidden, and available from the same menu: File, Genre in File and Searched As.

### What the Status column means

- **Ready to write**: a confident change was found. **Review**: a change was found but the match is weaker; double-click the row to check it.
- **Already correct**: the file already has the values that were found.
- **No genre found** or **No year found**: nothing to write for that field from the chosen source. "Ready to write (no genre found)" means the year will be written but there's no genre to add.
- **No match** or **No results**: the search found nothing usable. Use Edit Search.

### Fixing a search

CrateTag reduces DJ pool titles to the song name before searching. For example, `LET 'EM KNOW - KidCutUp Clap Intro (Dirty) - 11A - 84` searches as `LET 'EM KNOW`, and blends such as `Pop Dat Thang x Pop That` search the first song. The **Searched As** column shows exactly what was searched. If it's wrong, double-click it, type just the artist and song name, and it searches again. DJ names the cleanup misses can be added under Settings > Also ignore in titles.

## Updating to a new version

New versions are published to GitHub (github.com/amv617/CrateTag). When CrateTag opens, it checks there quietly; if a newer version exists, an **Update to …** button appears at the bottom right. Click it, or choose **CrateTag > Check for Updates…** any time. Confirm, and a Terminal window gets the new version from GitHub, rebuilds CrateTag.app, replaces the copy in Applications and reopens it. Your settings are kept. No download needed. (You can turn the automatic check off in Settings under Updates.)

If your CrateTag folder is connected to GitHub (it has a hidden .git folder), the update brings that folder up to date with GitHub too, so it always matches. Anything you changed by hand there is set aside safely first (git stash list shows it), and the previous version is kept in a branch called before-update.

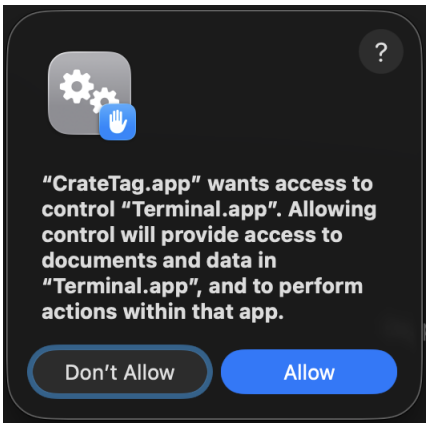
### Updating from a downloaded zip

This still works, for example without internet on the Mac or for a version sent to you directly.

1. Download the new **CrateTag.zip** into your Downloads folder. There's no need to unzip it (if your browser unzips it automatically, that works too).
2. In CrateTag, choose **CrateTag > Update from Downloads…**, or double-click **Update CrateTag.command** in your CrateTag folder.

3 **The first time only**, macOS asks: *"CrateTag.app" wants access to control "Terminal.app".* Click **Allow** (shown below). CrateTag uses this only to open Terminal and run the updater.

4 A Terminal window shows the progress. After about a minute the new version opens and the Terminal window closes by itself. If anything goes wrong, the window stays open so you can read the message.




The first-time prompt when updating from inside CrateTag. Click **Allow**.

**Clicked Don't Allow by mistake?** Open System Settings > Privacy & Security > **Automation**, find **CrateTag**, and turn on **Terminal**. Then choose Update from Downloads again. If CrateTag can't open Terminal, it shows a message with a button that opens that settings page.

**If macOS says "Update CrateTag.command" Not Opened** (Apple could not verify it), click **Done**, not Move to Trash. Files from a download are flagged by macOS. Clear the flag once with the commands below, then run the update again. From version 0.8.1 the updater and the build clear it for you, and CrateTag > Update from Downloads… isn't affected.

```
xattr -dr com.apple.quarantine ~/Downloads/CrateTag
cd ~/Downloads/CrateTag
./"Update CrateTag.command"
```

The updater can also be run from Terminal:

```
cd ~/Downloads/CrateTag
./"Update CrateTag.command"
```

To update by hand instead: quit CrateTag, unzip the new zip over your CrateTag folder, then run ./build_mac.sh --install (or ./run_from_source.sh if you don't use the app). The version shows in the window title (for example CrateTag v0.12.0) and under CrateTag > About CrateTag.

## Backups and undo

Settings > **Before writing** has three choices:

- **Keep an undo log** (default, recommended for large runs). Before each file is changed, its original year and genre are saved to a small file in ~/Music/CrateTag Backups. Thousands of tracks take only a few hundred KB. To restore, choose **Tracks > Undo a Previous Write…** and pick the log for that run.
- **Copy each file first.** A full copy of every MP3 goes into a dated folder, using as much disk space as the files themselves. CrateTag checks free space first and offers the undo log instead if there isn't room.

- **No backup.** Fastest, but a write can't be undone.

Only the year and genre tags are ever changed, so the undo log is all that's needed to put a track back the way it was. Undo restores the original text, including full dates or more than one genre.

# Good to know

- Keep Serato closed while writing. If your music is on an external drive, macOS may ask to allow access.
- **Only fill in empty fields** (Settings) never replaces a year or genre you've already set.
- Large batches take time: MusicBrainz allows about one request per second, so plan on roughly two to four seconds per track with both sources on. You can Stop at any point and analyze the rest later.
- Settings are stored in ~/Library/Application Support/CrateTag/settings.json. By default "Hip Hop" is written as "Hip-Hop"; that and other spellings can be changed there under genre_aliases.

## Activity log

CrateTag keeps a running record of what it does. Open it with **Help > Activity Log…** (Command-Option-L). Each line has the time and what happened:

- **Load**: which crate, folder or All Songs was opened, and how many tracks.
- **Analyze**: what was searched for each track and what was found (year, genre, match %, source), or why nothing was found.
- **Edit**: every change you type, shown as old → new, marked "not saved yet".
- **Write**: each file written with its old → new values, any failures with the reason, and a summary. If you click Write and nothing is written, the log says why (for example, the highlighted tracks had no changes).
- **Restore** from an undo log, setting changes, and any unexpected errors.

Type in the box at the top to filter (a file name, Write, FAILED). **Copy** puts what's shown on the clipboard, handy for sending to someone helping you. The log is saved in ~/Library/Logs/CrateTag/ and keeps itself to a few megabytes.

## Keyboard shortcuts

Every menu command can have a shortcut, and you can change any of them. Open **CrateTag > Settings…** (Command-comma) and click the **Keyboard Shortcuts** tab, or choose **Help > Keyboard Shortcuts…**.

1. Find the command (type in the search box to narrow the list).
2. Click its shortcut box and press the keys you want, for example Command-Shift-R.
3. To remove a shortcut, click the x in its box or press Delete while the box is selected.
4. Click **Save**. The new shortcut works right away.

If two commands would share the same keys, both turn red, a message says which ones, and Save stays greyed out until you change one. **Reset All to Defaults** puts every shortcut back.

| Command | Default shortcut |
| --- | --- |
| Reveal in Finder (all highlighted tracks) | Command-R |
| Analyze | Command-L |
| Write | Command-S |
| Set Year / Set Genre | Command-Y / Command-G |
| Highlight all tracks | Command-A |
| Search Tracks / Search Crates | Command-F / Command-Option-F |
| Remove highlighted from list (files stay on disk) | Command-Delete |
| Command palette (run any command by name) | Command-K |
| Show or hide Crates / Browse Folders | Command-Shift-C / Command-Shift-F |
| Add Folder / Add Files | Command-O / Command-Shift-O |
| Activity Log | Command-Option-L |
| Stop | Esc |

Commands without a default (Show Matches, Edit Search, Set Artist, Set Title, Check and Uncheck highlighted tracks, Undo a Previous Write and more) are listed too, so you can give them one. Reveal in Finder is also in the Tracks menu and when you right-click tracks.

## Troubleshooting

| Problem | Fix |
|---|---|
| command not found: brew | Run the installer command in section 1; it installs Homebrew when needed. |
| permission denied: ./run_from_source.sh | Run the installer command in section 1 once; it fixes permissions. |
| python3 shows 3.9 | That's Apple's old Python. The installer installs a newer one; open a new Terminal window afterward. |
| Installer: Homebrew couldn't be installed | Your Mac user needs to be an administrator. Or install Python from python.org, then run the installer again. |
| Discogs rejected the credentials | Generate a new token on discogs.com and paste it into Settings. |
| AcoustID: fpcalc not found | Run ./install_deps.sh --acoustid in your CrateTag folder. |
| Serato shows old values | Rescan ID3 tags for those tracks in Serato. |
| Need to undo a write | Tracks > Undo a Previous Write…, then choose that run's undo log. |
| Something didn't do what you expected | Open Help > Activity Log… to see exactly what CrateTag did and why. |
| Update or build failed | The Terminal window shows the last lines of the build log; the full log is in ~/Library/Logs/CrateTag/build.log. Your previous CrateTag.app is put back automatically. |
| Updater: no CrateTag download found | Make sure the new CrateTag.zip is in Downloads (not still downloading). |
| "Update CrateTag.command" Not Opened | Click Done, then run the three commands in Updating to a new version. |
| "CrateTag.app" wants access to control "Terminal.app" | Click Allow. This appears the first time you update from inside CrateTag. |
| The Dock or Finder still shows the old icon | macOS caches app icons. Quit CrateTag, drag it out of the Dock, then drag it back in from Applications. If Finder still shows the old one, restart the Mac. |
| Update from Downloads does nothing / says it can't open Terminal | System Settings > Privacy & Security > Automation > CrateTag > turn on Terminal. |